

Module-playerbots — Command Reference Manual

Complete CLI Task Dispatcher Specification & Operational Governance (Turtle WoW 1.18.1)

CLI REFERENCE MANUAL

TARGET REPOSITORY tortoise-wow-extended	TARGET BRANCH mantech-turtle	CLI DISPATCHER task.ps1 / tools/task.ps1	BUILD TOOLCHAIN Visual Studio 2022 x64 (MSVC)
PRIMARY UPSTREAM 1 cmangos/playerbots (master)	PRIMARY UPSTREAM 2 ileboii/core (vmangos-ike3-playerbots)	COMPILATION TARGET modules.lib (static)	LINKAGE TARGET mangosd.exe (Release x64)

1. EXECUTION MODES & DEPTH CONFIGURATION			
Mode Flag	Inspection Scope	Filtration & Triage Behavior	Execution Speed
FAST	Diff headers & metadata	Scans commit subjects and modified file paths. Detects explicit keywords (crash, assert, memory leak, UAF). Selects only high-confidence P0/P1 fixes.	Instantaneous (~1–2 sec)
NORMAL DEFAULT	Full watermark diffs & AST	Standard operational mode. Inspects full diffs, AST signatures, and Turtle WoW API compatibility. Triage candidates into P0 (Critical), P1 (Recommended), and P2 (Optional).	Fast (~5–10 sec in memory)
DEEP	Blame, callers & cross-donor	Exhaustive analysis. Evaluates cross-donor implementation differences between CMaNGOS and vMaNGOS. Traces call trees, checks semantic duplicates, and audits Dungeon Clear compatibility.	Deep trace (~20–40 sec)

2. TASK DISPATCHER CLI COMMANDS & SYNTAX			
Command & Category	Execution Syntax & Parameters	Functional Explanation & Workflow	Output / State Impact
status DIAGNOSTIC	<code>.\task.ps1 status</code> No additional parameters required.	Inspects and prints overall health across the target repository and donor checkouts. Reports active branch, commit HEAD SHA, working tree cleanliness, donor watermarks, unscanned commit counts, and durable ledger metrics.	Console report; read-only; no state modification.
scan DISCOVERY	<code>.\task.ps1 scan [source] [mode]</code> source: cmangos vmangos all mode: fast normal deep	Performs a non-destructive audit of all newly published upstream commits between the recorded watermark and donor HEAD. Evaluates changes against Turtle WoW 1.18.1 rules, classifies priorities, and records findings in the porting ledger. Batch auditing is permitted and recommended for scanning efficiency.	Updates source-watermarks.json and porting-ledger.json.
sync FULL PIPELINE	<code>.\task.ps1 sync [source] [mode]</code> source: cmangos vmangos all mode: fast normal deep	Executes the automated end-to-end backporting pipeline: updates donor references, audits candidate deltas in batch, then executes an atomic commit-by-commit loop . Each selected candidate is ported natively, compiled against modules.lib, linked into mangosd.exe, documented in a per-commit dossier, committed to target git, and immediately pushed to mantech-turtle. ADR-007: Batch commits and batch pushes are strictly prohibited.	Generates target git commits and remote pushes to mantech-turtle.
import-critical P0 TARGETED	<code>.\task.ps1 import-critical [source] [mode]</code> source: cmangos vmangos all	Filters donor backlog exclusively for high-severity P0 fixes: server crashes, null pointer exceptions, thread-safety deadlocks, memory corruption, and fatal bot provisioning failures. Ports and verifies only P0 candidates.	Atomic target commit & push per critical fix.
audit INVESTIGATION	<code>.\task.ps1 audit <source> <commit-sha></code> source: cmangos vmangos commit-sha: 8+ character SHA	Performs an isolated deep semantic audit of an individual upstream commit without touching target files. Checks whether the bug exists in target, maps file paths, evaluates Dungeon Clear impact, checks expansion compatibility, and writes a detailed audit report.	Generates JSON audit report in reports/commit-audits/.

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port MANUAL PORT	<pre>.\task.ps1 port <source> <commit-sha></pre> source: cmangos vmangos commit-sha: 8+ character SHA	Directly ports an explicitly requested upstream commit and its required dependencies into modules/mod-playerbots using target-native APIs. Runs the full compilation and linkage ladder, generates the dossier, and records the target commit SHA in the ledger.	Modifies module source; builds and verifies binaries.
verify-fast OPTION 1 DEFAULT	<pre>.\task.ps1 verify-fast</pre> Alias: verify-quick. Compiles modules.lib.	Triggers rapid compilation of affected static module code (modules.lib) using CMake parallel build (4 threads). Validates syntax, type signatures, template instantiations, and header inclusion in ~3 seconds with minimal token output (~3 lines).	Reports PASS/FAIL on modules.lib.
verify-full FULL GATE / OPTION 3	<pre>.\task.ps1 verify-full</pre> Compiles modules and links server daemon.	Compiles all modules and links the complete target server binary (mangosd.exe). Ensures there are zero unresolved external symbols, linker collisions, or missing virtual table entries across PlayerBots, Dungeon Clear, and core engine (~2–3 minutes).	Reports PASS/FAIL on mangosd.exe linkage.
verify-batch OPTION 2 BATCH	<pre>.\task.ps1 verify-batch</pre> Full compile & link pass for multi-commit batch.	Executes a single combined verification pass across a series of sequentially committed fixes. Compiles modules.lib then links mangosd.exe to verify the entire batch together, saving massive token output and turnaround latency.	Reports PASS/FAIL on batch modules and mangosd.
build-options INFORMATION	<pre>.\task.ps1 build-options</pre> Displays ADR-009 verification modes.	Summarizes the three authorized verification execution modes (Option 1: Fast Incremental, Option 2: Batch Verification, Option 3: Strict Full-Link) and provides guidance on token optimization and debugging workflows.	Console explanation of verification execution modes.
ledger REPORTING	<pre>.\task.ps1 ledger [source]</pre> source (optional): cmangos vmangos	Displays a structured tabular overview of all audited, ported, verified, duplicate, and rejected commits from state/porting-ledger.json. Shows commit SHA, author date, priority tier, triage status, and subject.	Console report; filters by donor if requested.
roadmap TRACKING	<pre>.\task.ps1 roadmap</pre> Displays milestone progress.	Outputs active milestone progress across project Phases 0 through 6 (Foundation, P0 Critical, Combat Parity, Group Coordination, RTSC/Travel, Polish, and Live Automation).	Console output of active milestone checklist.
widen-scope PROHIBITED GUARD	<pre>.\task.ps1 widen-scope</pre> Security & Stability Guard	Enforces ADR-006 : Unbounded historical scope widening beyond active watermarks is strictly prohibited. If triggered, the dispatcher halts immediately, explains the risks of scope explosion and regression hazards, and demands explicit double-confirmation.	Execution blocked; requires explicit operator double-confirmation.
commit-policy GOVERNANCE	<pre>.\task.ps1 commit-policy</pre> Prints binding architectural rules.	Displays the binding rules of ADR-007 : Batch auditing is permitted for scanning speed; batch commits and batch pushes are strictly prohibited. Every upstream commit ported must produce exactly 1 target commit and 1 remote push to mantech-turtle.	Console summary of commit granularity standards.
vanilla-mandate VANILLA INVARIANT	<pre>.\task.ps1 vanilla-mandate</pre> Prints ADR-008 expansion mandate.	Displays the binding rules of ADR-008 : Target is strictly Vanilla (Classic 1.12.1 / Turtle WoW 1.18.1 Classic+). Porting anything from TBC or WotLK is STRICTLY PROHIBITED . Only changes to Vanilla and Classic-related mechanics are permitted. All post-Vanilla commits are automatically disqualified as EXPANSION_INCOMPATIBLE.	Console summary of strict Vanilla exclusivity mandate.

3. BUILD & VERIFICATION EXECUTION MODES & TOKEN OPTIMIZATION (ADR-009)

Execution Mode	Per-Commit Step	Batch / Milestone Gate	Token & Time Cost	Debugging & Guarantees
Option 1: Fast Incremental RECOMMENDED	Compile modules.lib via .\task.ps1 verify-fast (~3 sec). Commit & push 1-by-1.	Full mangosd.exe link via .\task.ps1 verify-full at end of batch/phase.	~90% token reduction; ~3 sec latency per commit.	Compiler errors give exact file/line. Atomic commits preserve full git bisect.
Option 2: Batch Verification HIGH THROUGHPUT	Sequential git commits with atomic dossiers. Compilation deferred.	Single comprehensive pass: compile modules.lib + link mangosd.exe via verify-batch.	Lowest token cost (1 build pass per batch); fastest execution.	MSVC output pinpoints exact file/line on failure. Atomic commits preserve bisectability.
Option 3: Strict Full-Link P0 SAFETY	Compile modules.lib and link mangosd.exe after every single commit.	Verified independently on every commit before git push.	High token usage (hundreds of lines/commit); ~2–3 min per commit.	Instant per-commit link proof. Reserved for P0 threading, mutexes, and engine overhauls.

4. CANDIDATE TRIAGE & CLASSIFICATION HIERARCHY

Tier	Status Classification	Defect Class & Eligibility Criteria	Porting Action Policy
P0	CRITICAL	Crashes, segfaults, null pointer dereferences, assertions, memory corruption, UAF, infinite loops, bot provisioning deadlocks.	Immediate priority port. Port natively or with adaptation. Level 3 verification mandatory.
P1	RECOMMENDED	AI combat correctness, tank healing priority, threat handling, movement/pathing stalls, resurrect bugs, dispel/cure logic, group buffing.	Standard port candidate. Evaluated under NORMAL mode. 1-to-1 atomic commit & push.
P2	OPTIONAL / PENDING	Quality-of-life enhancements, chat shortcuts, inventory/food adjustments, bot PvP aggression scaling, travel distribution.	Port after P0/P1 stability baseline is established.
P3	DISQUALIFIED	TBC / WotLK expansion mechanics (STRICTLY PROHIBITED) , Arena teams, flying mounts, post-Vanilla raid scripts, cosmetic reformatting, duplicate code.	Disqualified and recorded as REJECTED or EXPANSION_INCOMPATIBLE in ledger.

5. ARCHITECTURAL GOVERNANCE & SAFETY INVARIANTS

1. Mandatory 1-to-1 Commit Granularity (ADR-007): Batch auditing is permitted for scanning efficiency. However, all target code modifications, verification builds, git commits, documentation dossiers, and remote pushes must occur strictly **commit-by-commit**. 1 Upstream Donor Commit = 1 Target Commit = 1 Remote Push to mantech-turtle. Squashing or batching donor commits into composite commits is strictly prohibited.

2. Strict Vanilla / Classic Exclusivity Mandate (ADR-008): This project is exclusively dedicated to **Vanilla / Classic WoW** (Classic 1.12.1 / Turtle WoW 1.18.1 Classic+). Porting **ANY** feature, spell, talent, combat rating, resilience mechanic, item, opcode, or strategy from The Burning Crusade (TBC 2.x), Wrath of the Lich King (WotLK 3.x), or later expansions is **STRICTLY PROHIBITED**. Only changes to Vanilla and Classic-related mechanics are permitted. Zero post-Vanilla pollution.

3. Scope Boundary Invariant & Double-Confirmation Gate (ADR-006): Widening the scan scope into historical donor backlogs beyond the active watermarks is strictly **PROHIBITED**. Scanning unbounded historical commits risks scope explosion, merge regressions, and TBC/WotLK pollution. The framework will never widen scope automatically; any expansion request requires an explicit operator warning and mandatory double-confirmation.

4. Level 3 Verification Gate & Turtle WoW Invariants: A module file compiling successfully is never sufficient proof of validity. mangosd.exe must link cleanly without unresolved symbols before any commit is marked verified. All ports must preserve Turtle WoW contracts: 11 playable races (Goblins & High Elves with custom starting zone spawn handling), 64-bit debuff streaming (sTWDebuff), custom entities (spells ≥ 40000, items ≥ 300000), and complete compatibility with modules/mod-dungeon-clear.